25901400 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCMBG203 - FOOD SAFETY, FOOD HYGIENE AND PUBLIC HEALTH

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Refrigeration helps to prevent food spoilage by: [K] / [CO1]

   a). Killing all microorganisms

   b). Increasing enzyme activity

   c). Slowing microbial growth

   d). Drying foods

2. Most common spoilage organisms of fruits and vegetables are: [U] / [CO1]

   a). Bacteria b). Virus

   c). Molds d). Protozoa

3. Which organism causes red spots on meat? [K] / [CO1]

   a). Pseudomonas b). Serratia

c). Bacillus d). Yeast

4. Ropiness in milk is mainly due to: [K] / [CO1]

a). Fat breakdown b). Slimy bacterial growth

c). Protein coagulation d). Putrefaction

5. Salmonellosis mainly affects: [U] / [CO2]

a). Digestive system b). Respiratory tract

c). Nervous system d). Circulatory system

6. Personal hygiene is important mainly to prevent: [U] / [CO2]

a). Chemical contamination b). Food spoilage only

c). Food-borne diseases d). Physical contamination

7. The causative organism of staphylococcal food poisoning is: [U] / [CO2]

a). Salmonella b). Clostridium botulinum

c). Staphylococcus aureus d). E. coli

8. Hair in food is classified as: [U] / [CO3]

a). Chemical hazard b). Biological hazard

c). Physical hazard d). Nutritional hazard

9. Which is NOT a chemical hazard? [K] / [CO3]

a). Cleaning agents b). Heavy metals

c). Viruses d). Pesticides

10. Packaging during transport helps to: [U] / [CO3]

a). Increase weight b). Prevent contamination

c). Improve taste d). Change color

[1x10 = 10]

Part B

Fill in the blanks questions

Answer all questions

Each question carries **1** marks

11. Flat sour spoilage mainly produces _____ acid. [U] / [CO1]

12. Cholera spreads through contaminated _____. [U] / [CO2]

13. Amoebiasis occurs by ingestion of contaminated food and _____. [K] / [CO2]

14. FSSAI ensures availability of safe and _____ food to consumers. [K] / [CO3]

15. Quaternary ammonium compounds are _____ sanitizing agents. [U] / [CO3]

[1x5 = 5]

Part C

True or false questions

Answer all questions

Each question carries **1** marks

16. Insect infestation is a major cause of cereal spoilage. [K] / [CO1]

17. ***Desulfotomaculum nigrificans*** is associated with sulfide stinker spoilage. [K] / [CO1]

18. GMP is concerned only with finished product testing. [K] / [CO3]

19. Raw or undercooked seafood is a common source of infection. [U] / [CO2]

20. Norovirus is a common cause of food-borne gastroenteritis. [K] / [CO2]

[1x5 = 5]

Part D

One word answer questions

Answer any **10** questions

Each question carries **1** marks

21. What is the main visible sign of hydrogen swell spoilage in cans? [K] / [CO1]

22. What is evisceration? [K] / [CO1]

23. Select one example of semi-perishable food. [U] / [CO1]

24. What does ISO 22000 standard relate to? [K] / [CO3]

25. Which type of food is commonly associated with *Bacillus cereus* poisoning? [K] / [CO2]

26. Select one mycotoxin produced by Aspergillus species. [K] / [CO2]

27. Select one preventive measure for bacterial food-borne diseases. [K] / [CO2]

28. Identify the strain of *E. coli* commonly causes food-borne gastroenteritis. [U] / [CO2]

29. Name one advantage of CIP over COP. [K] / [CO3]

30. List one critical control point in food processing. [K] / [CO3]

31. How does improper storage during transportation affect food safety? [K] / [CO3]

32. Is Giardia lamblia a protozoan or a helminth? [K] / [CO2]

[1x10 = 10]

Part E

Short answer questions

Answer any **4** questions

Each question carries **3** marks

33. List any three major types of spoilage in meat. [K] / [CO1]

34. Select the public health importance of typhoid fever. [U] / [CO2]

35. Explain the public health importance of Hepatitis A. [U] / [CO2]

36. Why are improperly canned foods a risk for botulism? [K] / [CO2]

37. Indicate any three sources of biological hazards in food. [U] / [CO3]

38. Explain how FAO and WHO work together to ensure food safety. [U] / [CO3]

[3x4 = 12]

Part F

Long answer questions

Answer any **1** questions

Each question carries **8** marks

39. Explain in detail the microbial causes of food spoilage with suitable examples. [U] / [CO1]

40. Define food-borne pathogens and explain their significance in public health. [K] / [CO2]

[8x1 = 8]